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System and method of low-waste multi-material resin printing

Abstract

A multi-material 3-D printing system and method including at least two printing heads each with a transparent window circumscribed by an ejection nozzle. Each ejection nozzle is coupled to a respective pump that pumps resin from a respective vat onto a respective window. The resin is cured from below the window by exposure to a digital image displayed by a micro display chip. To switch resins, the sample is moved across a plurality of suction nozzles towards a second printing head. A respective one of the suction heads is coupled to a vacuum that effectuates the intake of residual resin from the underside of the sample.

Inventors:	Xia; Chunguang (San Diego, CA)
Applicant:	BMF Nano Material Technology Co., Ltd (Shenzhen, CN)
Family ID:	1000008778419
Assignee:	BMF Nano Material Technology Co., Ltd (Shenzhen, CN)
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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
4575330	12/1985	Hull	N/A	N/A
5192559	12/1992	Hull	N/A	N/A
5933278	12/1998	Plummer	N/A	N/A
6011577	12/1999	Goodman	N/A	N/A
6607689	12/2002	Farnworth	N/A	N/A
7892474	12/2010	Shkolnik	N/A	N/A
8487822	12/2012	Jennings	N/A	N/A
9205601	12/2014	Desimone	N/A	N/A
9211678	12/2014	Desimone	N/A	N/A
9216546	12/2014	Desimone	N/A	N/A
9360757	12/2015	Desimone	N/A	N/A
9496675	12/2015	Plummer	N/A	N/A
11654617	12/2022	Xia et al.	N/A	N/A
11654619	12/2022	Xia	N/A	N/A
2005/0219400	12/2004	Poulsen	N/A	N/A
2005/0248061	12/2004	Shkolnik	N/A	N/A
2008/0174050	12/2007	Kikuchi	N/A	N/A
2009/0186306	12/2008	Haraldsson	N/A	N/A
2009/0196946	12/2008	Kihara et al.	N/A	N/A
2010/0291401	12/2009	Medina	N/A	N/A
2011/0101569	12/2010	Yasukochi	N/A	N/A
2011/0260365	12/2010	El-Siblani	N/A	N/A
2011/0285060	12/2010	Yamamoto	425/150	B33Y 40/00
2013/0021217	12/2012	Tsai	N/A	N/A
2013/0295212	12/2012	Chen	N/A	N/A
2015/0001763	12/2014	Elsey	N/A	N/A
2015/0247679	12/2014	Niimi	N/A	N/A
2017/0274586	12/2016	Hakkaku	N/A	N/A
2017/0334142	12/2016	Baldeck	N/A	N/A
2017/0355132	12/2016	Moore	N/A	B29C 35/02

2018/0056587	12/2017	Arai	N/A	N/A
2018/0056590	12/2017	Costabeber	N/A	N/A
2018/0194064	12/2017	Chen	N/A	N/A
2018/0200964	12/2017	Rockstroh	N/A	N/A
2019/0022947	12/2018	Ho et al.	N/A	N/A
2019/0111622	12/2018	Khalip	N/A	N/A
2020/0061917	12/2019	Osborn	N/A	B33Y 30/00
2020/0101665	12/2019	Beets	N/A	N/A
2020/0108465	12/2019	Sweetland	N/A	N/A
2021/0237345	12/2020	Xia et al.	N/A	N/A
2021/0276249	12/2020	Xia	N/A	N/A
2021/0331380	12/2020	Seo	N/A	N/A
2022/0048254	12/2021	Xia	N/A	N/A
2023/0398739	12/2022	Xia et al.	N/A	N/A
2024/0157642	12/2023	Korten	N/A	B33Y 30/00
2024/0198587	12/2023	Xia	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
102928910	12/2012	CN	N/A
104827667	12/2014	CN	N/A
104991353	12/2014	CN	N/A
105390056	12/2015	CN	N/A
206718470	12/2016	CN	N/A
107756814	12/2017	CN	N/A
107756814	12/2017	CN	N/A
108437448	12/2017	CN	N/A
108437448	12/2017	CN	N/A
109466061	12/2018	CN	N/A
109483872	12/2018	CN	N/A
109483872	12/2018	CN	N/A
111873431	12/2019	CN	N/A
111873433	12/2019	CN	N/A
112549537	12/2020	CN	N/A
113927897	12/2021	CN	N/A
1732746	12/2010	EP	N/A
3107703	12/2019	EP	N/A
4232262	12/2022	EP	N/A
3894184	12/2023	EP	N/A
4326529	12/2023	EP	N/A
4384379	12/2023	EP	N/A
H07329188	12/1994	JP	N/A
2004314406	12/2003	JP	N/A
2005219400	12/2004	JP	N/A
2009083240	12/2008	JP	N/A
I568571	12/2016	TW	N/A
2008069564	12/2007	WO	N/A
2009042671	12/2008	WO	N/A
2012051598	12/2011	WO	N/A
2014042486	12/2013	WO	N/A

2016115095	12/2015	WO	N/A
2017157851	12/2016	WO	N/A
2019014098	12/2018	WO	N/A
2020181620	12/2019	WO	N/A
2021056720	12/2020	WO	N/A
2021103502	12/2020	WO	N/A
2021165878	12/2020	WO	N/A

OTHER PUBLICATIONS

“Three-dimensional fabrication of heterogeneous microstructures using soft membrane deformation and optofluidic maskless lithography”, Seung Ah Lee, Su Eun Chung, Wook Park, Sung Hoon Lee and Sunghoon Kwon, Lab Chip, 2009, 9, 1670-1675. cited by applicant

B. J. Lei, et al., “A wideband, pressure-driven, liquid-tunable frequency selective surface,” IEEE Microwave Wireless Components Letters, vol. 21, No. 9, pp. 465-467, 2011. cited by applicant

Jingi Wang et al., “Liquid metal-based plasmonics” Optics Express, vol. 20, No. 2, p. 2346-2353, Jan. 2012. cited by applicant

K. Ikuta and K. Hirowatari, “Real three dimensional micro fabrication using stereo lithography and metal molding,” 6th IEEE Workshop on Micro Electrical Mechanical Systems, 1993. cited by applicant

S. Maruo and K. Ikuta, “Three-dimensional microfabrication by use of single-photon-absorbed polymerization,” Appl. Phys. Lett., vol. 76, 2000. cited by applicant

S. Maruo and S. Kawata, “Two-Photon-Absorbed Near-Infrared Photopolymerization for Three dimensional Microfabrication,” J. MEMS, vol. 7, pp. 411, 1998. cited by applicant

S. Kawata, H. B. Sun, T. Tanaka and K. Takada, “Finer features for functional microdevices,” Nature, vol. 412, pp. 697, 2001. cited by applicant

Bertsch et al., “Microstereolithography using a liquid crystal display as dynamic mask-generator”, Microsystem Technologies, p. 42-47, 1997. cited by applicant

Beluze et al., “Microstereolithography: a new process to build complex 3D objects , Symposium on Design, Test and microfabrication of MEMs/MOEMSs”, Proceedings of SPIE, v3680, n2, p. 808-817, 1999. cited by applicant

Kowsari, et al., “High-efficiency high-resolution multimaterial fabrication for digital light processing-based three-dimensional printing,” 3D Printing and Additive Manufacturing 5(3):185-193, 2018. cited by applicant

Han et al., “Rapid multi-material 3D printing with projection micro-stereolithography using dynamic fluidic control,” Additive Manufacturing 27(11):606-615, 2019. cited by applicant

T. Takagi and N. Nakajima, 4th International Symposium on Micro Machine and Human Science, 1993. cited by applicant

S. Maruo, O. Nakamura and S. Kawata, “Three-dimensional microfabrication with two-photon-absorbed photopolymerization,” Optics Letters, vol. 22, pp. 132, 1997. cited by applicant

S. Maruo and K. Ikuta, “Movable microstructures made by two-photon three-dimensional microfabrication,” in Proceedings of 1999 International Symposium on Micromechatronics and Human Science, 1999, pp. 173-178. cited by applicant

Arnaud Bertsch, Sébastien Jiguet, Paul Bernhard and Philippe Renaud, “Microstereolithography: a Review,” Materials research society symposium proceedings, vol. 758, 2003, p. 3-15. cited by applicant

Notice of Allowance dated Jan. 31, 2025 for U.S. Appl. No. 17/312,173. cited by applicant

Panasonic (“Micro Laser Distance Sensor,” in public at least in Apr. 2014, available at <https://www.farnell.com/datasheets/1863584.pdf>). (Year: 2014). cited by applicant

International Search Report and Written Opinion for PCT Patent Application No.

PCT/IB2023/051584, 10 pages. cited by applicant

International Search Report and Written Opinion for PCT Patent Application No. PCT/IB2021/059731 dated Feb. 28, 2022. cited by applicant
International Search Report and Written Opinion for PCT Patent Application No. PCT/IB2022/053395 dated Jun. 27, 2022, 12 pages. cited by applicant
International Search Report and Written Opinion for PCT Patent Application No. PCT/IB2021/059731 dated Feb. 28, 2022, 22 pages. cited by applicant
International Search Report and Written Opinion for PCT Patent Application No. PCT/IB2022/057569 dated Nov. 8, 2022, 16 pages. cited by applicant
Non-Final Office Action for U.S. Appl. No. 18/033,405 dated Jan. 3, 2025. cited by applicant

Primary Examiner: Robinson; Michael M.

Attorney, Agent or Firm: Young Basile Hanlon & MacFarlane, P.C.

Background/Summary

FIELD

(1) The present disclosure relates to additive manufacturing, and more particularly, to systems and methods for high-resolution 3-D printing over a large area while using multiple printing materials.

BACKGROUND

(2) Stereolithography was originally conceived as a rapid prototyping technology used to create true-scale models of production components directly from computer aided design (CAD) in a rapid (faster than before) manner. Since its conception, and through its disclosure in U.S. Pat. No. 4,575,330, stereolithography has greatly aided engineers and designers in visualizing complex three-dimensional part geometries, detecting errors in prototype schematics, testing critical components, and verifying theoretical designs at relatively low costs and improved time frames.

(3) During the past decades, continuous improvements in the field of micro-electromechanical systems (MEMS) have led to the emergence of micro-stereolithography (μ SL), which inherits basic principles from traditional stereolithography but with much higher spatial resolution. See, e.g., Ikuta et al., “Real three dimensional micro fabrication using stereo lithography and metal molding,” Proceedings of MEMS '93, 6th IEEE Workshop on Micro Electro Mechanical Systems, San Diego, CA, Jan. 25-28, 1993, pp. 42-47.

(4) The resolution of μ SL was improved with the development of single-photon polymerization and two-photon polymerization techniques to achieve printed features of less than 200 nm. See, e.g., Maruo et al., “Three-dimensional microfabrication by use of single-photon-absorbed polymerization,” Applied Physics Letters 76(19):2656-2658, 2000; Maruo et al., “Two-photon-absorbed near-infrared photopolymerization for three-dimensional microfabrication,” Journal of Microelectromechanical Systems 7(4):411-415, 1998; and Kawata et al., “Finer features for functional microdevices—micromachines can be created with higher resolution using two-photon absorption” Nature 412(6848):697-698, 2001.

(5) The speed of μ SL was dramatically increased with the development of projection micro-stereolithography (P μ SL). See, e.g., Bertsch et al., “Microstereolithography using liquid crystal display as dynamic mask-generator,” Microsystem Technologies, 3(2):42-47, 1997; and Beluze et al., “Microstereolithography: A New Process to Build Complex 3D Objects,” Symposium on Design, Test and Microfabrication of MEMs/MOEMs,” Proceedings of SPIE 3680(2):808-817, 1999. The core of this technology is a high resolution spatial light modulator, which is either a liquid crystal display (LCD) panel or a digital light processing (DLP) panel, each of which are available from micro-display industries.

(6) The display size of a DLP chip is currently limited to about 13×13 mm. Therefore, when the projected pixel size is same as the physical pixel size (e.g., $5\times 5\text{ }\mu\text{m}$ to $8\times 8\text{ }\mu\text{m}$), a single-exposure area is limited to 13×13 mm. To print a larger area with single exposure, the size of the projected pixels must be increased. However, this reduces the printing resolution and may therefore be undesirable.

(7) In some cases it is desirable to print individual models from multiple materials, for example from materials having different optical, mechanical, electrical, or chemical properties. P μ SL printing materials comprise light-curable resins and composites. The terms “resin,” “material,” and “printing material” may be used interchangeably herein.

(8) Multi-material printing via P μ SL can be problematic because switching materials during P μ SL processes can add significant delays. To overcome some of these delays, methods like coat-then-jet-clean and print-then-flush were developed. See, e.g., respectively, Kowsari, et al., “High-efficiency high-resolution multimaterial fabrication for digital light processing-based three-dimensional printing,” *3D Printing and Additive Manufacturing* 5(3):185-193, 2018; and Han et al., “Rapid multi-material 3D printing with projection micro-stereolithography using dynamic fluidic control,” *Additive Manufacturing* 27(11):606-615, 2019. However, issues with air bubbles and excessive consumption of resin have hindered widespread adoption of these methods.

(9) During P μ SL printing, a resin layer is defined between a vat (or reservoir) of resin and the sample stage. A “sample” may refer to the 3-D model as it is being printed layer by layer, and a “sample stage” may refer to the most recently printed layer of the sample. There are generally three methods for defining the resin layer in P μ SL. The first method uses a free surface where the layer thickness is defined by a distance between the resin's free surface and a sample stage. However, with this method it may take more than a half an hour to define a $10\text{ }\mu\text{m}$ thick resin layer having a viscosity of 50 cP. This is because of the slow viscous motion of resins. The second and third methods for defining the resin layer in P μ SL use a transparent membrane or a hard window, respectively. However, with these methods there is currently no good way to define a $10\text{ }\mu\text{m}$ thick (or thinner) resin layer over an area of 5×5 cm (or larger). Thus, while these methods may enable defining a resin layer faster than the free-surface method, they remain impractically slow.

Additionally, during hard-window P μ SL printing, the fluidic dynamic force created as the sample approaches the window to define the thin resin layer before exposure (or during separation from the window after exposure) may be strong enough to damage the sample.

(10) The disclosed embodiments solve several of the above-described problems and other problems in the prior art.

SUMMARY

(11) The several embodiments provide for fast and precise P μ SL printing over a large area with a defined print-layer thickness. For example, the print area may be 10×10 cm and the print-layer thickness may be $10\text{ }\mu\text{m}$. A sample is printed layer-by-layer, where the printing of a given layer may be referred to as a “scan” whether it is printed in one or more exposures. In some embodiments there may be two or more printing heads each immersed in and covered by a respective printing material (e.g., a respective resin). The resin-coated printing heads may be utilized in sequentially in alternation to print a multi-material sample.

(12) A printing head may be frustoconical and comprise an optically transparent flat tip at its narrow end. The tip may comprise gas permeable materials, including oxygen-permeable materials such as Polydimethylsiloxane (PDMS) and Teflon AF (from DuPont). The tip may be as large as (or larger than) the exposure area of a DLP chip or it may be a fraction thereof.

(13) There may be at least one printing head for each different printing material (e.g., for each different resin). Each printing head may be disposed within a vat of its assigned resin, under the free surface of the resin by approximately 1-3 mm. During a scan, a lens projects an image onto a selected one of the printing heads. In some embodiments the lens is disposed gravitationally below the printing head. The lens focuses an image upwards to the transparent flat tip of the printing head

to optically expose the resin that is disposed between the tip and the bottom of the sample. The exposed resin cures to form at least a portion of a newly added layer of cured material to the underside of the sample. In some embodiments the entire tip comprises a transparent window and in other embodiments only a portion of the tip comprises a transparent window.

(14) In some embodiments there may be a plurality of lenses, for example one lens for each of a plurality of printing heads. In these embodiments, the image projected through each lens may be gated by a shutter to prevent the image from reaching any non-selected printing heads during a given exposure. A shutter may optically precede the lens or it may optically follow the lens, i.e., the shutter may be positioned at the input or output of the lens, respectively.

(15) In some embodiments each printing head is surrounded by a ring-shaped ejection nozzle. The nozzle is connected to a liquid pump that pumps resin from a reservoir to the tip of the printing head. The flow of resin from the nozzle pushes the free surface of the resin above the tip by approximately 1-3 mm to form a pool of resin above the tip. This fountain-like design enables the tip and/or the pool of resin to be elevated above the free surface of the resin within the vat (or conversely, the free surface of the resin within the vat may be lower than the tip and/or the pool of resin). A lower level of resin in the vat (lower free surface) means that less resin may be present in the vat during the printing process, which may result in less wasted resin. To be clear, the fountain-like design enables subsequent layers of a sample to be printed without the entire bottom surface of the sample being in contact with the free surface of the resin within the vat.

(16) In some embodiments there may be two parallel suction nozzles disposed between two vats of resin, where a first suction nozzle is closer to a first vat of first resin and a second suction nozzle is closer to a second vat of second resin. Each suction nozzle may have a gas-flow velocity of approximately 1-10 m/s driven by a vacuum of approximately 0.2-1.0 atm of pressure difference. When a sample translates from the first vat of a first resin to the second vat of a second resin, the wet bottom surface of the sample moves past (slides above) the suction nozzles by approximately 0.5-1.0 mm, causing any residual uncured (liquid) first resin to be sucked into and collected by the first suction nozzle. The collected first resin may be pumped back to the designated printing head to be used for printing a subsequent layer of the sample. When the sample translates in the reverse direction, the wet bottom surface of the sample again slides above the suction nozzles and any residual uncured (liquid) second resin is sucked into and collected by the second suction nozzle. Embodiments that use three or more resins may comprise three or more suction nozzles that collect residual resin in a similar manner.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) A more complete appreciation of the present disclosure and many of the attendant advantages thereof will be readily obtained as the same becomes better understood by reference to the following detailed description when considered in connection with the accompanying drawings.

(2) FIG. 1 shows a schematic of a low-waste multi-material PμSL system.

(3) FIG. 2 shows a schematic of a printing head for a PμSL system.

(4) FIGS. 3A-3E show alignments of exposures during stitch printing.

(5) FIGS. 4A-4C show single-, stitch-, and array-exposure printing modes, respectively.

(6) FIG. 5 shows a schematic of a vacuum nozzle between vats of resin.

(7) FIGS. 6A-6C show a schematic of a low-waste multi-material PμSL system at various stages during material switching.

DETAILED DESCRIPTION

(8) The present disclosure may be understood more readily by reference to the following detailed description of the disclosure taken in connection with the accompanying drawing figures, which

form a part of this disclosure. It is to be understood that this disclosure is not limited to the specific devices, methods, conditions, or parameters described and/or shown herein, and that the terminology used herein is for the purpose of describing particular embodiments by way of example only and is not intended to be limiting of the claimed disclosure. Also, as used in the specification and including the appended claims, the singular forms “a,” “an,” and “the” include the plural, and reference to a particular numerical value includes at least that particular value, unless the context clearly dictates otherwise.

(9) The following numerals are used to describe various features of the embodiments. **5** printing apparatus **10** sample **20** sample platform **30a,b** first and second vat **40a,b** first and second storage reservoir **50a,b** first and second resin **60a,b** first and second printing head **70a,b** first and second tip **80a,b** first and second gas-permeable window **90a,b** first and second ring-shaped nozzle **100a,b** first and second resin pool **110a,b** first and second liquid pump **120a,b** first and second suction nozzle **130a,b** first and second suction reservoir **140a,b** first and second valves **150a,b** first and second lens **160a,b** first and second shutter **170** vacuum chamber **180** manifold **190** computer **200** DLP and light source **210** beam splitter **220** mirror **230** XYZ stage assembly **240** charge-coupled device (CCD) **250a,b** first and second tall blade **260a,b** first and second short blade

(10) FIG. 1 shows an embodiment of a multi-material resin printing apparatus **5** for printing a 3-D sample **10** on the underside of a sample platform **20**. The sample platform **20** may be coupled to a precision XYZ stage assembly **230** that can move the sample platform **20** laterally in the X and Y directions and vertically in the Z direction relative to each of a first printing head **60a** and a second printing head **60b** (e.g., three precision stages adapted to move the sample **10**, affixed to a sample platform **20**). In some embodiments, relative motion between the printing heads **60** and the sample platform **20** may be enabled by coupling each stage of the XYZ stage assembly **230** to various parts of the printing apparatus **5** such as the printing heads **60**.

(11) Each printing head **60** may be disposed in a respective vat **30** and partially or completely submerged in and/or coated by a respective resin **50**. Each vat **30** is above and fluidically coupled to a respective storage reservoir **40** such that liquid resin **50** in the vat **30** may drain into the storage reservoir **40** (e.g., each storage reservoir **40** is disposed below a respective vat **30** and fluidically coupled thereto via an orifice). The vat **30** may collect liquid resin **50** as it drips off of a sample **10** during printing.

(12) A multi-material printing process may begin by generating (or otherwise obtaining) a digital 3-D model via a computer **190**. The 3-D model may be an assembly of multiple models each representing different materials. The 3-D model may be sliced, or divided, into layers approximately 5-20 μm thick. Each layer may be represented by one or more 2-D digital images, where any given digital image corresponds to a single material. In other words, if a given layer comprises two materials, then there would be at least two digital images for that layer.

(13) A sample **10** may be printed in layers, one at a time, by projecting and focusing a series of optical images onto a selected one of the printing heads **60**. To print a given layer, the computer **190** may send a digital image to a DLP and light source **200** where it may be transformed into an optical image (e.g., the DLP is adapted to display a digital image received from the computer **190**). The DLP and light source **200** may project the optical image towards a selected one of the printing heads **60**, for example, towards a first printing head **60a** for printing a first resin **50a** or towards a second printing head **60b** for printing a second resin **50b**. Each printing head **60a** or **60b** may be selected (or deselected) by opening (or closing) a respective shutter **160a** or **160b** (e.g., each respective shutter **160a** or **160b** is adapted to selectively prohibit the digital image from reaching a respective lens **150a** or **150b**). A lens **150a** or **150b** may be disposed between each respective shutter **160a** or **160b** and the respective printing head **60a** or **60b** to focus the optical image(s) (e.g., the lens **150a** or **150b** may be disposed below a respective gas-permeable window **80** and adapted to focus the digital image).

(14) In some embodiments the optical image projected by the DLP and light source **200** may

encounter a beam splitter **210** that partially reflects the optical image towards the first shutter **160a** and partially transmits the optical image therethrough towards a mirror **220** that reflects the optical image towards the second shutter **160b** (e.g., the mirror **220** is adapted to reflect the partially transmitted digital image towards the second shutter **160b**). In some embodiments a charge coupled device (CCD) **240** may be aligned with an optical axis of the first lens **150a** to monitor the projection of the optical image onto the first printing head **60a** (e.g., adapted to monitor the projection of the optical image). The CCD **240** may be positioned on a side of the beam splitter **210** opposite the first shutter **160a**.

(15) As shown in FIG. 2, a printing head **60** may be frustoconical with a flat tip **70** at the frustum (e.g., the printing head **60** may define an upward-facing frustum). The tip **70** may comprise a transparent gas-permeable window **80** (e.g., a flat, sealed, optically transparent, and gas-permeable window **80**) through which an optical image may be transmitted. The gas-permeable window **80** may have lateral dimensions to accommodate an entire optical image projected by the DLP and light source **200**. For example, a DLP and light source **200** that utilizes a DLP chip having 1920×1080 resolution with 10×10 μm pixels may project an optical image with a 22 mm diagonal dimension (via Pythagorean's theorem). Thus, a circular gas-permeable window **80** may be chosen to have a diameter of 25 mm.

(16) The gas-permeable window **80** may be referred to as a membrane or a film, and it may be constructed from or coated by any suitable material, preferably those with superior optical clarity such as DuPont Teflon AF2400 or polydimethylsiloxane (PDMS). Gas permeability helps to prevent the gas-permeable window **80** from sticking to resin **50** that cures during photo polymerization due to exposure to an optical image. Oxygen permeability is especially advantageous because oxygen inhibits photo-crosslinking. The gas-permeable window **80** may have a thickness of 130 μm.

(17) Each printing head **60** may be circumscribed by a ring-shaped nozzle **90**, e.g., an ejection nozzle having an annular (or polygonal annular) opening, that is fluidically coupled to a respective liquid pump **110** capable of pumping resin **50** from a storage reservoir **40** to the tip **70** of the printing head **60** (e.g., each liquid pump **110** is adapted to pump resin **50** from a storage reservoir **40** to a printing head **60**). The ring-shaped nozzle **90** may be frustoconical (or frusto-pyramidal) with an outer wall extending upwards and inwards at an angle of approximately 50 degrees from the horizontal, and the ring-shaped nozzle **90** may have an annular opening adjacent to and circumscribing the tip **70** of the printing head **60** with a gap of approximately 500 μm. Resin **50** may be pumped by the liquid pump **110** into and through the ring-shaped nozzle **90**. As the resin **50** exits the annular opening of the ring-shaped nozzle **90**, it may flow towards the center of the gas-permeable window **80** due to the inward angle of the outer wall of the ring-shaped nozzle **90**, thereby creating a resin pool **100** that covers the top surface of the gas-permeable window **80**. The thickness of the resin pool **100** may be controlled by adjusting the flow rate of the liquid pump **110**. In some embodiments a thickness of 1-3 mm may be preferred. The liquid pump may be any suitable type, for example a non-contact peristaltic pump or a diaphragm pump, and it may pump at a flow rate of approximately 5-10 ml/s. During printing, excess resin **50** of the resin pool **100** may flow off the tip **70** into a respective vat **30** and reservoir **40**. The terms “frustoconical” and “frusto-pyramidal” may be used interchangeably herein, and the terms “annular” and “polygonal annular” may be used interchangeably herein.

(18) An optical image that is focused on a printing head **60** creates a focused image at or adjacent to the gas-permeable window **80**, and therefore, at or adjacent to the resin pool **100** coating the top surface of the gas-permeable window **80**. The bright areas of the optical image cause the resin **50** of the resin pool **100** to cure, or polymerize, whereas the dark areas of the optical image do not cause curing or polymerization.

(19) Due to size limitations of currently available LCD and DLP chips, a single exposure may not be large enough to print an entire layer (of a given material) of a sample **10**. For example, a DLP

chip having a pixel resolution of 1920×1080 resolution with $10 \times 10 \mu\text{m}$ pixels creates an optical image of $19.2 \times 10.8 \text{ mm}$. If a cross-section of the sample **10** is larger than $19.2 \times 10.8 \text{ mm}$, then the corresponding layer must be printed with multiple exposures, i.e., in multiple sections.

(20) In some embodiments multiple-exposure printing may be achieved by dividing a digital image corresponding to an oversized layer into multiple digital sub-images. Each sub-image corresponds to an optical image no larger than the maximum projected image of the DLP or LCD chip at the desired pixel size. For example, a printing apparatus **5** that utilizes a DLP chip having a pixel resolution of 1920×1080 may need to print an oversized layer having a pixel resolution of 3800×2000 . In this case, the computer **190** may generate (or otherwise obtain) four digital sub-images for that layer, each having a pixel resolution of 1900×1000 . The printing apparatus **5** may then print the oversized layer one section at a time, where each section corresponds to one of the sub-images.

(21) Adjacent edges of neighboring sections may be printed to overlap one another by $5\text{-}20 \mu\text{m}$, as indicated by FIGS. **3B-3C**, to improve mechanical strength of the resulting layer at the section boundaries. The precise positions and amounts of overlap at section boundaries may be accurately controlled by the XY stage of the XYZ stage assembly **230**. However, misalignment elsewhere in the printing apparatus **5** may cause errors in overlapping of section boundaries. As indicated in FIG. **1**, there are at least two frames of reference for each printing head **60**. A first frame of reference corresponds to alignment between the DLP and light source **200** and the printing head **60**, and a second frame of reference corresponds to alignment between the printing head **60** and the sample platform **20**. If these two frames of reference are misaligned with each other, then errors in overlapping of section boundaries like those shown in FIGS. **3D-3E** may result. Specifically, FIG. **3D** represents misalignment in the Y direction and FIG. **3E** represents misalignment in the X direction.

(22) A typical error requirement for PμSL printing may be $10 \mu\text{m}$. However, it is not uncommon for an XY stage to have an offset error that is greater than this amount, and further, the offset may be nonlinear with respect to travel distance. Therefore, to counter misalignment errors during multiple-exposure printing, offsets may be measured at a plurality of points distributed in both the X and Y directions of a maximally sized layer. In some embodiments five or more uniformly distributed points may be measured in a maximally sized, square-shaped layer. The measurements may be fit to a second order (or greater) polynomial that may be used to compensate for the observed nonlinear offsets of the XY stage.

(23) FIGS. **4A-4C** shows three printing modes enabled by the XY stage of the XYZ stage assembly **230**. In each figure, the maximum layer size is represented by the dotted outer rectangle and the optical-image exposure size is represented by the dashed inner rectangles (nine exposures per layer). FIG. **4A** shows “single-exposure” printing, where the printed area fits within a single exposure. In this mode, the XY stages of the XYZ stage assembly **230** do not need to move provided that only one resin **50** is used. FIG. **4B** shows “stitch-exposure” printing, where a large image is divided into sub-images each of which is independently printed. In some embodiments each exposure may overlap neighboring exposures by $5\text{-}20 \mu\text{m}$ as described earlier (overlaps are not shown in FIG. **4B**). FIG. **4C** shows “array-exposure” printing, where the same small image is printed independently multiple times. For low-volume production, array-exposure printing may be faster than single-exposure printing. Array-exposure and stitch-exposure printing may be combined, for example when multiple identical samples need to be printed and each sample has at least one layer that cannot be printed with a single exposure.

(24) For array-exposure and stitch-exposure printing in both X and Y directions, it may be advantageous to laterally size the vat **30** beneath each printing head **60** to be at least four times larger than the maximum size of a printed layer, e.g., at least four times larger than the sample platform **20**. This prevents residual resin **50** from dripping from the sample **10** and/or sample platform **20** outside the vat **30**. In other words, the vat **30** should be at least twice as long as the

sample platform **20** in the X direction and at least twice as long in the Y direction. In some embodiments the printing head **60** may be moved while the sample **10** is kept stationary, and in other embodiments both the sample **10** and the printing head may be moved.

(25) When the printing apparatus **5** finishes printing a first layer, the Z stage of the XYZ stage assembly **230** may move the sample platform **20** (and sample **10**) up vertically by at least a layer thickness to define a next layer of liquid resin **50** for printing. However, it may be advantageous to move the sample **10** away from the printing head **60** in the X and/or Y directions (laterally) before moving the sample **10** in the vertical direction. This is because the shear fluid force that results between the printing head **60** and the sample **10** when moving the sample laterally is generally much less than the normal fluid force that results between the printing head **60** and the sample **10** when moving the sample **10** vertically. For example, the vacuum pressure (normal force per unit area) caused by separating two surfaces wetted by resin commonly used in PμSL printing can be described by the following equation: $\sigma = -pI + 2\mu\epsilon$, where σ is the fluid stress tensor, p is the pressure, I is the identity tensor, μ is the fluid viscosity and ϵ is velocity gradient tensor (or fluid strain tensor). A resin viscosity $\mu = 50$ cP and a velocity $\epsilon = 10$ mm/s yields a vacuum pressure on the order of $1E5$ Pa. In contrast, the same two wetted surfaces sliding past each other with a gap of $20\ \mu\text{m}$ yields a shear pressure of $1E2$ Pa—approximately three orders of magnitude lower than the normal vacuum pressure. Thus, separating the sample **10** from the printing head **60** laterally instead of vertically helps to prevent damaging or deforming delicate 3-D printed features of the sample **10**. As indicated above, the vat **30** should be at least twice as long at the sample platform **20** in the X direction and at least twice as long in the Y direction (for array or stitch printing in both dimensions) to prevent residual resin **50** from dripping from the sample **10** and/or sample platform **20** outside the vat **30**.

(26) If the printing apparatus **5** needs to switch between resins **50** during multi-material printing, the XY stage of the XYZ stage assembly **230** may move the sample **10** (on the underside of the sample platform **20**) between printing heads **60**. However, the underside of the sample **10** may be coated with residual first resin **50a** that could drip into a second vat **30b** meant only for a second resin **50b**. To prevent such cross-contamination, the sample **10** may be moved over (slid across) a first suction nozzle **120a** to remove any residual resin **50a**. Similarly, to prevent cross-contamination in the opposite direction, e.g., when switching from resin **50b** to resin **50a**, the sample **10** may be moved over a second suction nozzle **120b** to remove any residual resin **50b**. Indeed, the printing apparatus may include one suction nozzle **120** for each different resin **50**.

(27) FIG. 5 shows two suction nozzles **120a** and **120b** disposed between the first vat **30a** and the second vat **30b**, where the first suction nozzle **120a** may be closer to the first vat **30a** and the second suction nozzle **120b** may be closer to the second vat **30b**. In some embodiments an outer wall of each suction nozzle **120a** and **120b** may contact or be shared with a respective vat **30a** and **30b**. Each suction nozzle **120a** and **120b** may be coupled to a respective suction reservoir **130a** and **130b** that may in turn be coupled to a vacuum chamber **170** (or any other vacuum source including a vacuum pump) via a manifold **180**. Each suction reservoir **130a** and **130b** may be connected to a respective storage reservoir **40a** and **40b** via a respective valve **140a** and **140b**.

(28) Consider the case when the printing apparatus **5** switches from a first resin **50a** to a second resin **50b**. As the sample **10** moves away from the first vat **30a** it encounters the first suction nozzle **120a**. At this time the first valve **140a** is closed and the manifold **180** has diverted the vacuum of the vacuum chamber **170** to only the first suction reservoir **130a**. In so doing, air (or gas) is sucked into the upwards-facing opening of the first suction nozzle **120a** as indicated by the dashed arrows in FIG. 5. This rapid air intake sucks residual resin **50a** from the sample **10**. As the sample **10** continues to move towards the second vat **30b**, no resin **50a** is sucked in by the second suction nozzle **120b** because there is no vacuum in the second suction reservoir **130b** (because the manifold **180** has diverted the vacuum to only the first suction reservoir **130a**). The sample **10** may continue moving until it reaches the second printing head **60b**. At this time the manifold **180** may divert the

vacuum away from the first suction reservoir **120a** and the first valve **140a** may be opened to recirculate any captured residual resin **50a** into the first storage reservoir **50a**. Now consider the case when the printing apparatus **5** switches back from the second resin **50b** to the first resin **50a**. When the sample **10** encounters the second suction nozzle **120a**, the second valve **140b** is closed and vacuum has been diverted by the manifold **180** to only the second suction reservoir **130b**. The second suction nozzle **120b** therefore sucks residual resin **50b** from the sample **10** while the first suction nozzle **120a** does not.

(29) The side walls (or rim) of each suction nozzle **120** may have different heights. Consider the first suction nozzle **120a**. The wall closer to the vat **30a** may define a first tall blade **250a** and the wall further from the vat **30a** may define a first short blade **260a**. The first short blade **260a** allows greater air flow than the first tall blade **250a** as the sample **10** moves past the first suction nozzle **120a**. Thus, the dominant flow of air is in a direction towards the first vat **30a** and away from the second vat **30b**, e.g., opposite the direction of motion of the sample **10**. This helps to prevent resin **50a** from splashing into or otherwise entering the second suction nozzle **120b** and thereby contaminating resin **50b**. In some embodiments a gap between the two suction nozzles **120** is open to external air to allow unhindered air supply, which may be important if the sample **10** spans across both suction nozzles **120** during movement of the sample **10** past the suction nozzles **120**.

(30) In some embodiments the difference in height between a tall blade **250** and a short blade **260** may be approximately 1 mm. In some embodiments each suction nozzle **120** may have an opening of approximately 0.5-1.0 mm, resulting in an air (or gas) flow velocity of approximately 1-10 m/s when driven by a pressure difference from the vacuum chamber **170** of approximately 0.2-1.0 atm. In some embodiments the sample **10** may move laterally past (slide above) a suction nozzle **120** at approximately 5 mm/s or less and at a vertical distance of approximately 0.5-1.0 mm. A higher air (or gas) flow rate and a slower movement of the sample **10** may result in more residual resin **50** being sucked from the sample **10** per unit time, especially for resins **50** with viscosities greater than approximately 500 cP.

(31) FIGS. **6A-6C** show an exemplary multi-material printing sequence for the printing apparatus **5**. FIG. **6A** indicates an optical image being projected through an open first shutter **160a**, through a first lens **150a**, and onto the first printing head **60a** adjacent to the sample **10**. Excess resin **50a** may be collected by a first vat **30a**. FIG. **6B** shows the sample **10** moving away from the first printing head **60a** in preparation for printing with a second resin **50b**. The sample **10** moves laterally past a first suction nozzle **120a**, which sucks residual resin **50a** from the sample **10**. FIG. **6C** indicates an optical image being projected through an open second shutter **160b**, through a second lens **150b**, and onto the second printing head **60b** adjacent to the sample **10**. Excess resin **50b** may be collected by a second vat **30b**.

(32) While several embodiments of the disclosure have been shown in the drawings, it is not intended that the disclosure be limited thereto, as it is intended that the disclosure be as broad in scope as the art will allow and that the specification be read likewise. Therefore, the above description should not be construed as limiting, but merely as exemplifications of particular embodiments.

Claims

1. A 3-D printing system, comprising: at least one micro display chip coupled to a light source and adapted to display a digital image received from a computer; at least two printing heads, each disposed below a sample, above a respective one of at least two collection vats, and defining an upward-facing frustum comprising a flat, sealed, optically transparent, and gas-permeable window circumscribed by an ejection nozzle fluidically coupled to a respective collection vat; at least two lenses, each disposed below a respective window and adapted to focus the digital image onto a layer of resin coating a top surface of the respective window; at least two suction nozzles, each

disposed below the sample, adjacent to a respective collection vat, fluidically coupled to the respective collection vat, and fluidically couplable to a vacuum source, wherein application of a vacuum to a respective suction nozzle effectuates a suction at an upward-facing opening thereof; and three precision stages adapted to move the sample, affixed to a sample platform, laterally (X and/or Y) and vertically (Z) relative to the at least two printing heads and laterally (X and/or Y) relative to the at least two suction nozzles.

2. The system of claim 1 wherein the opening of each suction nozzle is circumscribed by a rim having a tall edge opposite a short edge, the tall edge being adjacent to an individual one of the at least two collection vats to which the suction nozzle is fluidically coupled.

3. The system of claim 1 further comprising at least two suction reservoirs, each fluidically coupled to a respective suction nozzle and to a manifold that is fluidically coupled to the vacuum source, wherein the manifold diverts the application of the vacuum to a respective suction reservoir one at a time.

4. The system of claim 3 further comprising at least two storage reservoirs and at least two fluid pumps adapted to pump resin from a respective storage reservoir to a respective printing head, wherein each storage reservoir is disposed below a respective collection vat and fluidically coupled thereto via an orifice, and each storage reservoir is disposed adjacent to a respective suction reservoir and fluidically coupled thereto via a valve.

5. The system of claim 1 further comprising: at least two shutters, each adapted to selectively prohibit the digital image from reaching a respective lens; a beam splitter adapted to partially reflect the digital image towards a first shutter and to partially transmit the digital image towards a mirror as a partially transmitted digital image; and a mirror adapted to reflect the partially transmitted digital image towards a second shutter.

6. The system of claim 5 further comprising a charge coupled device (CCD) adapted to monitor the digital image focused by a respective one of the lenses, wherein the CCD is disposed on a side of the beam splitter opposite the respective lens.

7. The system of claim 1 wherein each collection vat is at least twice as long as the sample platform in a first lateral direction and at least twice as long as the sample platform in a second lateral direction.
